CME 307 / MS&E 311 / OIT 676: Optimization

LP geometry, modeling and solution techniques

Professor Udell

Management Science and Engineering
Stanford

September 29, 2026

First quiz Thursday next week

- Take paper when you arrive
- Write your SUNetID[1] and your name.
- We will project the question.
- You will have 5 minutes to write your answer on the paper.
- When time is up, put your pen or pencil down and stop writing.

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since a 5 minute quiz is very short, we will grade on a **check** basis.

- **no check:** no serious attempt
- **check**: a serious attempt
- **check+**: correct or nearly correct

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Exit ticket

Please fill out the exit ticket (linked from course schedule spreadsheet) before you leave:

- Helps us improve the course

We will not grade this, but we will read it!



Course overview

the practice of optimization consists of:

- modeling (real-world problems $\rightarrow$ optimization problems)
- analysis (properties of optimization problems)
- algorithms (how to solve optimization problems)

all three are necessary for success in optimization!

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- last class: mostly modeling
- today: mostly analysis
- next class: algorithms and some more modeling

Outline

LP standard form

LP inequality form

What kinds of points can be optimal?

Solving LPs

Modeling

Linear programming: standard form

standard form linear program (LP)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & x \geq 0 \end{array}$$

optimal value $p^\star$, solution $x^\star$ (if it exists)

- any x with $Ax = b$ and $x \geq 0$ is called a **feasible point**
- if problem is infeasible, we say $p^\star = \infty$
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Q: why? how to check?

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Q: why? how to check?
A: otherwise infeasible or redundant rows; use gaussian elimination to check and remove

LP example: diet problem

We an planning a backpacking trip, and want to minimize the total weight of the food packed subject to nutritional requirements. We have a list of essential nutrients and how much an active person needs of each. We also know the weight of each food, and how much of each nutrient is in each food.

- x_j servings of food j, $j = 1, \ldots, n$
- c_j weight per serving
- a_{ij} amount of nutrient i in food j
- b_i required amount of nutrient i, $i = 1, \ldots, m$

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- define the **cone** generated by $A = [A_1, \ldots A_n]$:

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Q: how many columns are needed to represent a feasible $b \in \mathbb{R}^m$?

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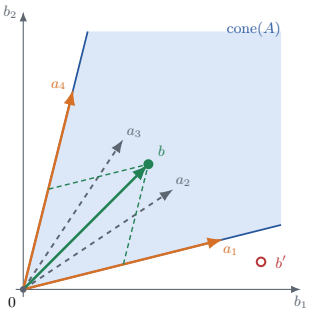
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Q: how many columns are needed to represent a feasible $b \in \mathbb{R}^m$?
A: at most m (here, b uses only a_1 and a_4)



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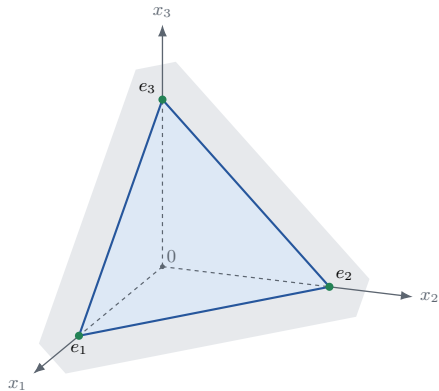
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- LP is feasible if hyperplane $\{x \mid Ax = b\}$ intersects the positive orthant

example: in $\mathbb{R}^3$, $\mathbf{1}^T x = 1$ cuts the orthant in the triangle $\mathbf{conv}\{e_1, e_2, e_3\}$



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- the feasible set $\{x : Ax = b,\ x \geq 0\}$ is convex

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LP inequality form

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another useful form for LP is **inequality form**

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interpretation: halfspaces

- $a_i^T x \leq b_i$ defines a **halfspace**
- $Ax \leq b$ defines a **polyhedron**: intersection of halfspaces
- LP is feasible if polyhedron $\{x \mid Ax \leq b\}$ is nonempty

LP example: production planning

- x_i units of product i
- c_i cost per unit
- a_{ij} amount of resource j used by product i
- b_j amount of resource j available
- d_i demand for product i

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  $c^T x + f^T z$, $z_i \in \{0, 1\}$, $x_i \leq M z_i$ for M large

LP inequality form to standard form

standard form to inequality form

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inequality form to standard form

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \leq b \end{array} \quad \rightarrow \quad \begin{array}{ll} \text{minimize} & c^T (x_+ - x_-) \\ \text{subject to} & A(x_+ - x_-) + s = b \\ & s, x_+, x_- \geq 0 \end{array}$$

so both forms have the same expressive power, and feasible sets are polyhedra

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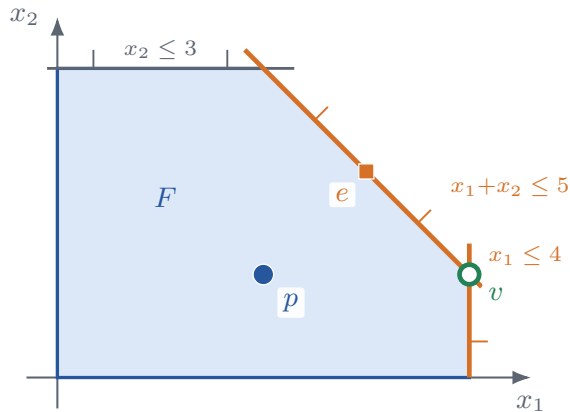
active variables. for nonnegative variable $x \geq 0$, variable i is **active** if $x_i > 0$
example: active slack variables are dual to active constraints

$$\begin{array}{rcl} Ax \leq b & \iff & Ax + s = b,\ s \geq 0 \\ a_i^T x = b_i & \iff & s_i = 0 \\ \text{constraint } i \text{ is active} & \iff & \text{slack variable } s_i \text{ is inactive} \end{array}$$

Active constraints: example

$$F = \left\{ x \in \mathbb{R}^2 \;\middle|\; \begin{array}{l} x \geq 0,\ x_1 \leq 4, \\ x_2 \leq 3,\ x_1 + x_2 \leq 5 \end{array} \right\}$$

- corner $v = (4, 1)$: $x_1 \leq 4$ and $x_1 + x_2 \leq 5$ active
- edge point $e = (3, 2)$: only $x_1 + x_2 \leq 5$ active
- interior point $p = (2, 1)$: none active



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Q: Does there always exist an extreme solution?

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proof: x is a vertex of S. suppose its defining vector is c and consider the optimization problem

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Vertices

define **vertex**: $x \in S \subset \mathbb{R}^n$ is a vertex of set S if for some vector $c \in \mathbb{R}^n$,

$$c^T x < c^T y \quad \forall y \in S \setminus \{x\}$$

interpretation: $\{z : c^T z = c^T x\}$ is a hyperplane that intersects S only at x.
we say this hyperplane **supports** S at x

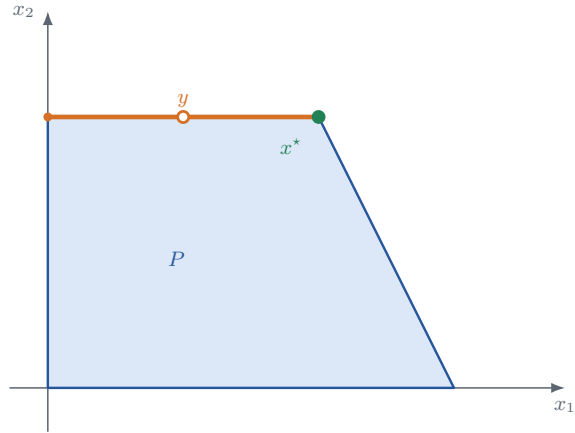
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x is the unique optimum of this problem, so the proof of this statement follows from the previous proof.

Extreme points and supporting hyperplanes

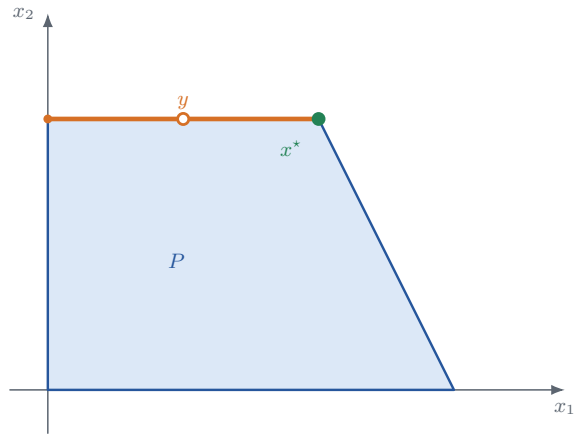
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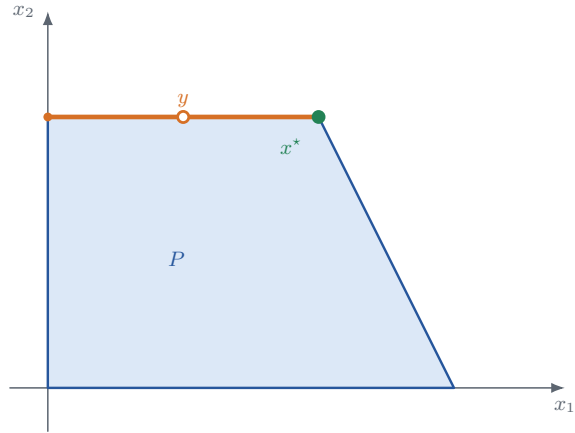
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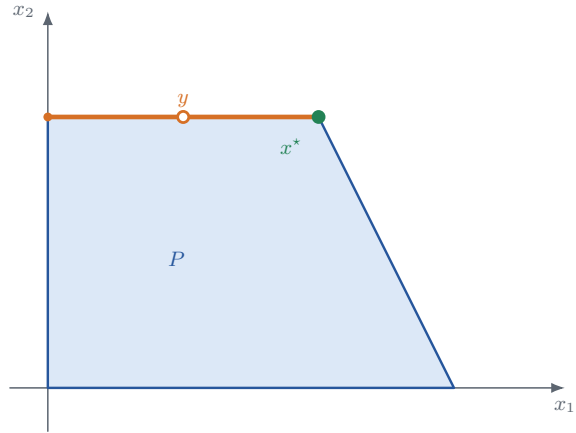
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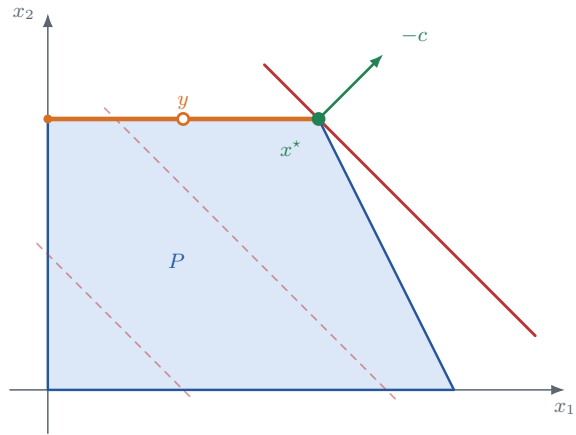
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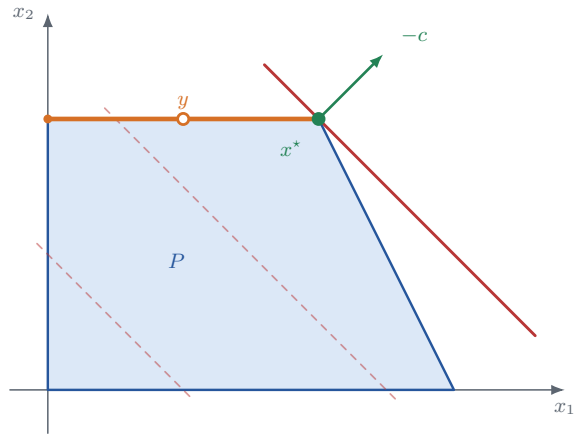
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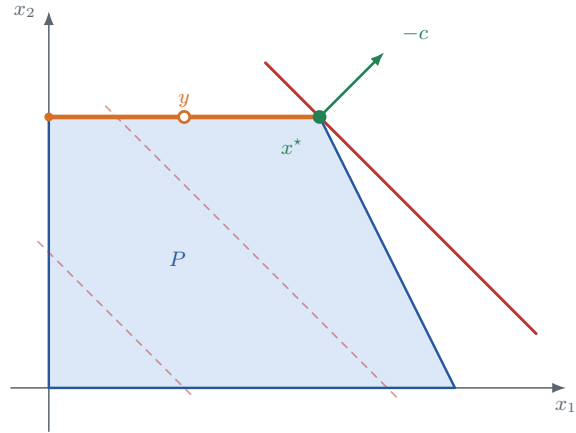
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recall the standard form LP

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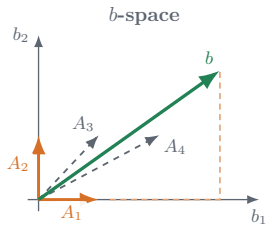
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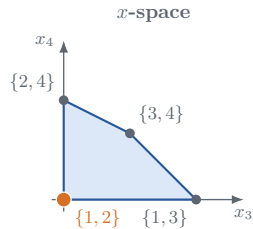
Q: how to find a BFS?

A: choose m linearly independent columns of A and set $x = A_S^{-1} b$; check $x \geq 0$.

A basis produces a basic feasible solution

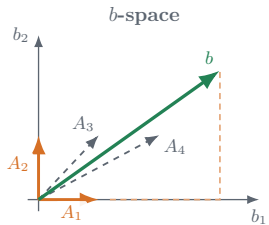
$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

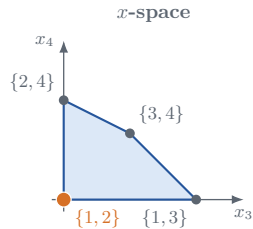




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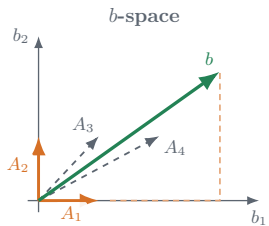


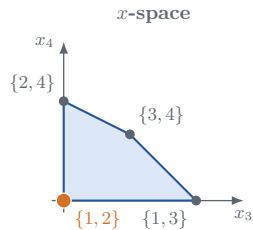


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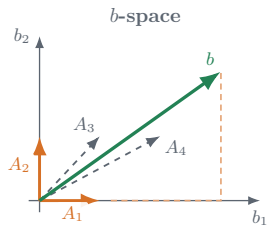


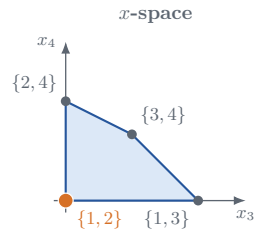


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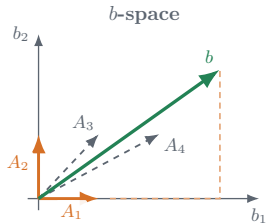


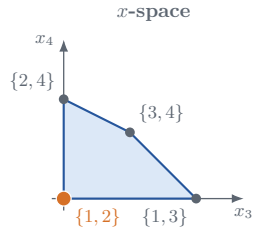


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Q: is $S = \{1, 4\}$ a basis? does it give a BFS?
A: a basis, but $x = (-1, 0, 0, 2) \not\geq 0$: basic but **infeasible**

Extreme point $\iff$ vertex $\iff$ BFS

fact. consider the feasible set in standard form $F = \{x \mid Ax = b,\ x \geq 0\}$ in $\mathbb{R}^n$. the following are equivalent:

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implications: since any polyhedron $Ax \leq b$ can be rewritten in standard form,

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we have already shown that vertex $\implies$ extreme point. need to show

- extreme point $\implies$ BFS
- BFS $\implies$ vertex

Extreme point $\implies$ BFS

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1. The columns in A_S are linearly independent. Otherwise, there is a nonzero $d \in \mathbb{R}^n$, supported on S, with $Ad = 0$. Since $x_i^\star > 0$ on S, for sufficiently small $\epsilon > 0$,

$$x^+ = x^\star + \epsilon d, \qquad x^- = x^\star - \epsilon d$$

are nonnegative. They also satisfy $Ax^+ = Ax^- = b$. Hence $x^+, x^- \in F$, but

$$x^\star = \tfrac{1}{2}x^+ + \tfrac{1}{2}x^-, \qquad x^+ \neq x^-,$$

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2. Complete A_S to a basis. Since A has rank m, extend S to B with $|B| = m$ and A_B invertible. Then $x_{\bar{B}}^\star = 0$, and $Ax^\star = b$ gives

$$x_B^\star = A_B^{-1} b.$$

Thus $x^\star$ is a BFS.

BFS $\implies$ vertex

suppose $x^\star$ is a BFS of F with active set S and A_S invertible. define $c \in \mathbb{R}^n$ as

$$c_i = \begin{cases} 0 & \text{if } i \in S \\ 1 & \text{otherwise} \end{cases}$$

so $c^T x^\star = 0$.

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- $x^\star$ is the only point in F supported on S, as $\mathrm{nullspace}(A_S) = 0$,
- so any other feasible point $x \in F$ has a positive objective value $c^T x > 0$.

hence $x^\star$ is a vertex of F with defining vector c.

Outline

LP standard form

LP inequality form

What kinds of points can be optimal?

Solving LPs

Modeling

Solving LPs

algorithms:

- enumerate all vertices and check
- fourier-motzkin elimination
- simplex method
- ellipsoid method
- interior point methods
- first-order methods
- ...

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remarks:

- enumeration and elimination are simple but not practical
- simplex was the first practical algorithm; still used today
- ellipsoid method is the first polynomial-time algorithm; not practical
- interior point methods are polynomial-time and practical
- first-order methods are practical and scale to large problems

Example of Fourier-Motzkin elimination

consider the system of inequalities

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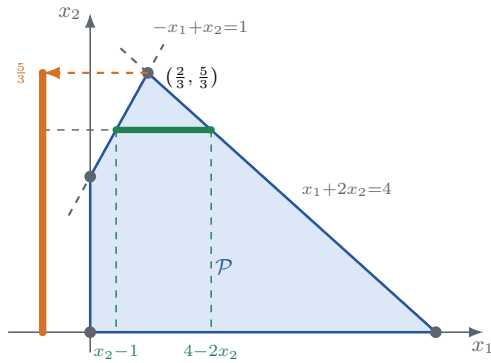
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$\implies x_2 \in [0, 5/3]$.
elimination method also shows projection of a polyhedron is a (closed) polyhedron



Enumerate vertices of LP

can generate all extreme points of LP: for each $S \subseteq \{1, \ldots, n\}$ with $|S| = m$,

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n choose m is $\binom{n}{m} = \frac{n!}{m!(n-m)!}$ ("exponentially many")

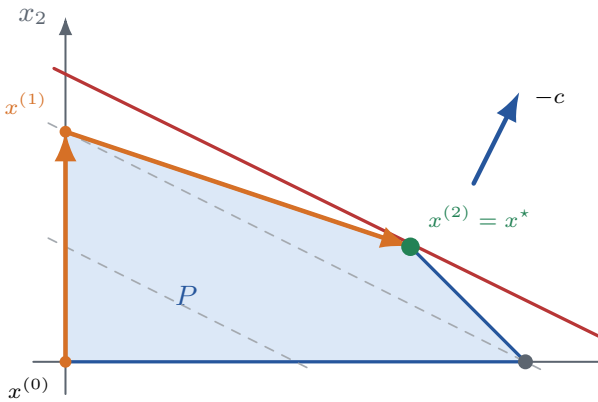
Simplex algorithm: example

$$
\begin{array}{ll}
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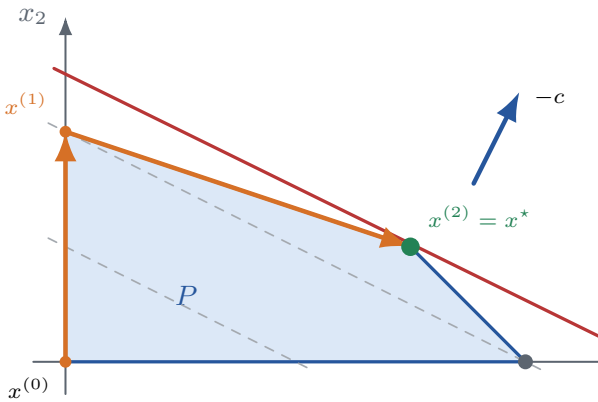
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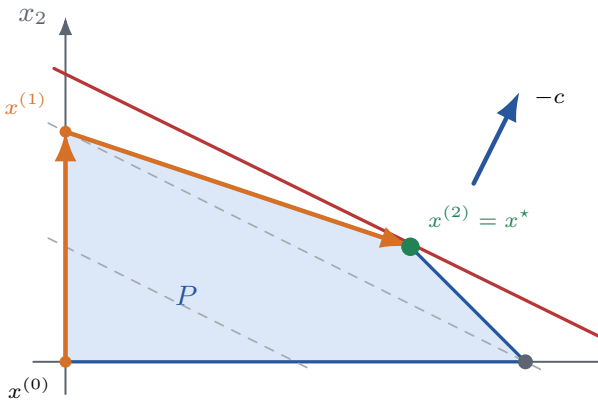
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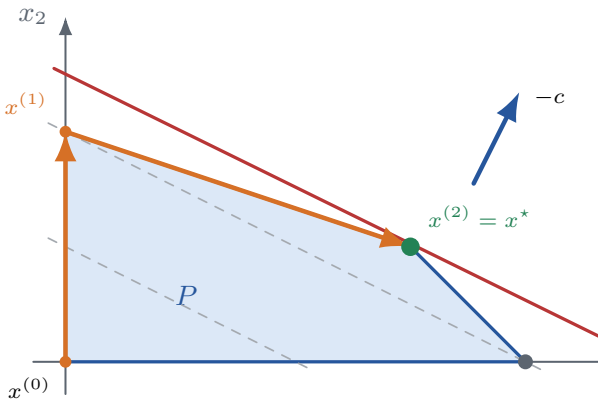
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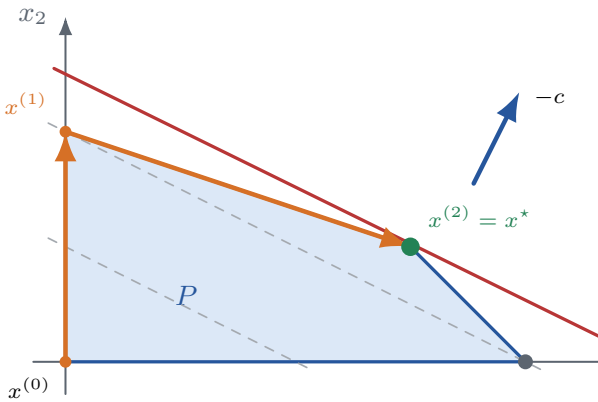


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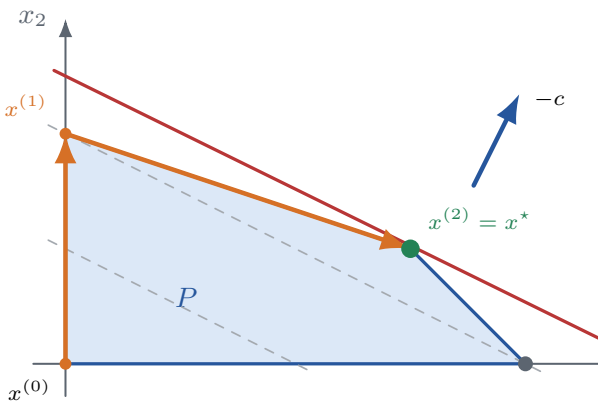


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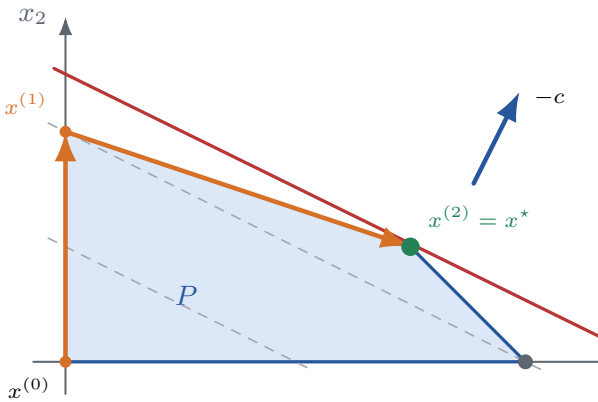


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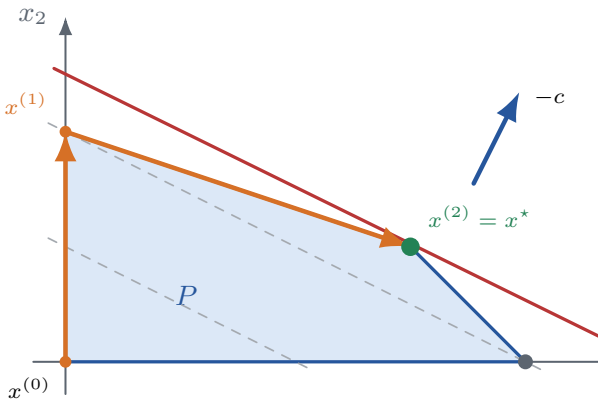


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Q: why stop at $x^{(2)}$?
A: no neighboring vertex improves the objective



Simplex algorithm

basic idea: local search on the vertices of the feasible set

- start at BFS x and evaluate objective $c^T x$
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discuss in groups:

- how to find an initial BFS?
- how to find a neighboring BFS with better objective?
- how to prove optimality?

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- $(x, z) = (x, 0)$ is a BFS of this problem $\iff$ x is a BFS of the original problem

Find a better neighboring BFS

start with BFS x with active set S, $x_S > 0$. (called a **non-degenerate** BFS.)
construct the j**th basic direction** d^j by turning on variable $j \notin S$

$$x^+ \leftarrow x + \theta d^j, \qquad \text{stepsize } \theta > 0$$

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- how does objective change if we move to $x^+ = x + \theta d^j$?

$$c^T x^+ - c^T x = \theta c^T d^j = \theta(c_j - c_S^T A_S^{-1} A_j)$$

Reduced cost

define **reduced cost** $\bar{c}_j = c_j - c_S^T A_S^{-1} A_j$, $j \notin S$

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let's show the first fact.

if $\bar{c} \geq 0$, x is optimal

let x be a BFS with basis S, and let x' be any feasible point. Since

$$A_S x_S = b, \qquad A_S x'_S + A_{\bar{S}} x'_{\bar{S}} = b,$$

we have

$$x'_S = x_S - A_S^{-1} A_{\bar{S}} x'_{\bar{S}}.$$

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because $\bar{c}_{\bar{S}} \geq 0$ and $x'_{\bar{S}} \geq 0$. therefore no feasible point has lower objective value than x, so x is optimal.

Outline

LP standard form

LP inequality form

What kinds of points can be optimal?

Solving LPs

Modeling

LP solvers

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  - power systems
    https://jump.dev/JuMP.jl/stable/tutorials/applications/power_systems/

Oro Verde case + tutorial

https://github.com/stanford-cme-307/demos/tree/main/gurobipy

A modeling workflow

an optimization model describes exactly which decisions are allowed

1. define each decision variable and its bounds
2. translate one requirement at a time
3. test the formulation on boundary cases

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three recurring ideas:

- introduce an auxiliary variable to expose a linear relation
- use binary variables to encode yes/no decisions
- use bounds to turn conditional constraints on and off

Two standard-form reformulations

feature	new variables	substitution
$a^T x \leq b$ in standard form	$s \geq 0$	$a^T x + s = b$
$x_i \in \mathbb{R}$ in standard form	$x_i^+, x_i^- \geq 0$	$x_i = x_i^+ - x_i^-$

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substitute these identities everywhere, including the objective

$$c_i x_i \quad \longmapsto \quad c_i x_i^+ - c_i x_i^- .$$

The epigraph trick

the **epigraph** of f contains every point above its graph:

$$\operatorname{epi} f = \{(x, t) : f(x) \leq t\}.$$

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minimizing the upper bound t makes it tight:

$$\min_x \sum_i |x_i| \quad \Longleftrightarrow \quad \begin{array}{ll} \text{minimize}_{x,t} & \mathbf{1}^T t \\ \text{subject to} & -t \leq x \leq t, \end{array}$$

$$\min_x \max_i f_i(x) \quad \Longleftrightarrow \quad \begin{array}{ll} \text{minimize}_{x,t} & t \\ \text{subject to} & f_i(x) \leq t, \quad \forall i. \end{array}$$

add the original constraints to both sides; the right-hand problems are LPs when each f_i is affine

Exercise: worst-case residual

reformulate as an LP:

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Each constraint gives $|a_i^T x - b_i| \leq t$; the objective finds the smallest uniform bound.

Binary variables encode propositions

let $z_A, z_B \in \{0, 1\}$ record whether projects A and B are funded

$$\text{“if } A\text{, then } B\text{”} \qquad \Longleftrightarrow \qquad z_A \leq z_B$$

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the inequality excludes exactly the forbidden truth assignment; binary variables make the model a mixed-integer LP

Counting binary decisions

for a binary variable, $1 - z_i$ encodes "not i"; sums count how many propositions are true:

$$\begin{aligned} \text{at least one} &: \sum_i z_i \geq 1, \\ \text{at most one} &: \sum_i z_i \leq 1, \\ \text{exactly } k &: \sum_i z_i = k. \end{aligned}$$

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an implication between binary variables is also a counting constraint:

$$z_A = 1 \implies z_B = 1 \qquad \iff \qquad z_A \leq z_B$$

Exercise: project logic

write linear constraints for the following rules:

1. fund A only if E is funded
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Q: Why is $z_A + z_B \leq z_H$ too strong?
A: It forbids funding both A and B, even when H is funded.

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M must cover the largest possible violation of $a^T x \leq b$

Bounds determine a valid big-M

if x already lies in a known set X, the tightest nonnegative choice is

$$M = \max\left\{0,\ \max_{x \in X}(a^T x - b)\right\}.$$

Bounds determine a valid big-M

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if p_i is fixed data, preprocess data to remove ineligible assignments; no need for big-M

Exercise: a plant that can be on or off

the decisions are production $x \geq 0$ and operating status $z \in \{0, 1\}$.

the data are:

- $0 < L \leq U$: minimum and maximum production when the plant operates
- $c \geq 0$: production cost per unit
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Q: Model the feasible production levels and the production cost.
A:

$$Lz \leq x \leq Uz, \qquad \text{cost} = cx + fz.$$

If $z = 0$, then $x = 0$; if $z = 1$, then $L \leq x \leq U$.

An equivalence needs two implications

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for continuous q, the reverse implication introduces a strict boundary; use a nonzero tolerance, or allow either value of z at $q = k$

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Q: Find two linear constraints that define z.
A: Set $q = \sum_{i=1}^{5} x_i$, $U = 5$, and $k = 3$:

$$\sum_{i=1}^{5} x_i \geq 3z, \qquad \sum_{i=1}^{5} x_i \leq 2 + 3z.$$

The first implication forces the count up when $z = 1$; the second forces $z = 1$ when the count reaches three.

Logical modeling: the short version

- encode propositions with binary variables

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solver indicator constraints express implications directly, but good variable bounds still matter

Exit ticket

Please fill out the exit ticket (linked from course schedule spreadsheet) before you leave:

- Helps us improve the course

We will not grade this, but we will read it!



Upcoming deadlines

- **Thursday, October 1:** Quiz 1
- **Friday, October 2:** Homework 1 due

see the course schedule for links and submission details